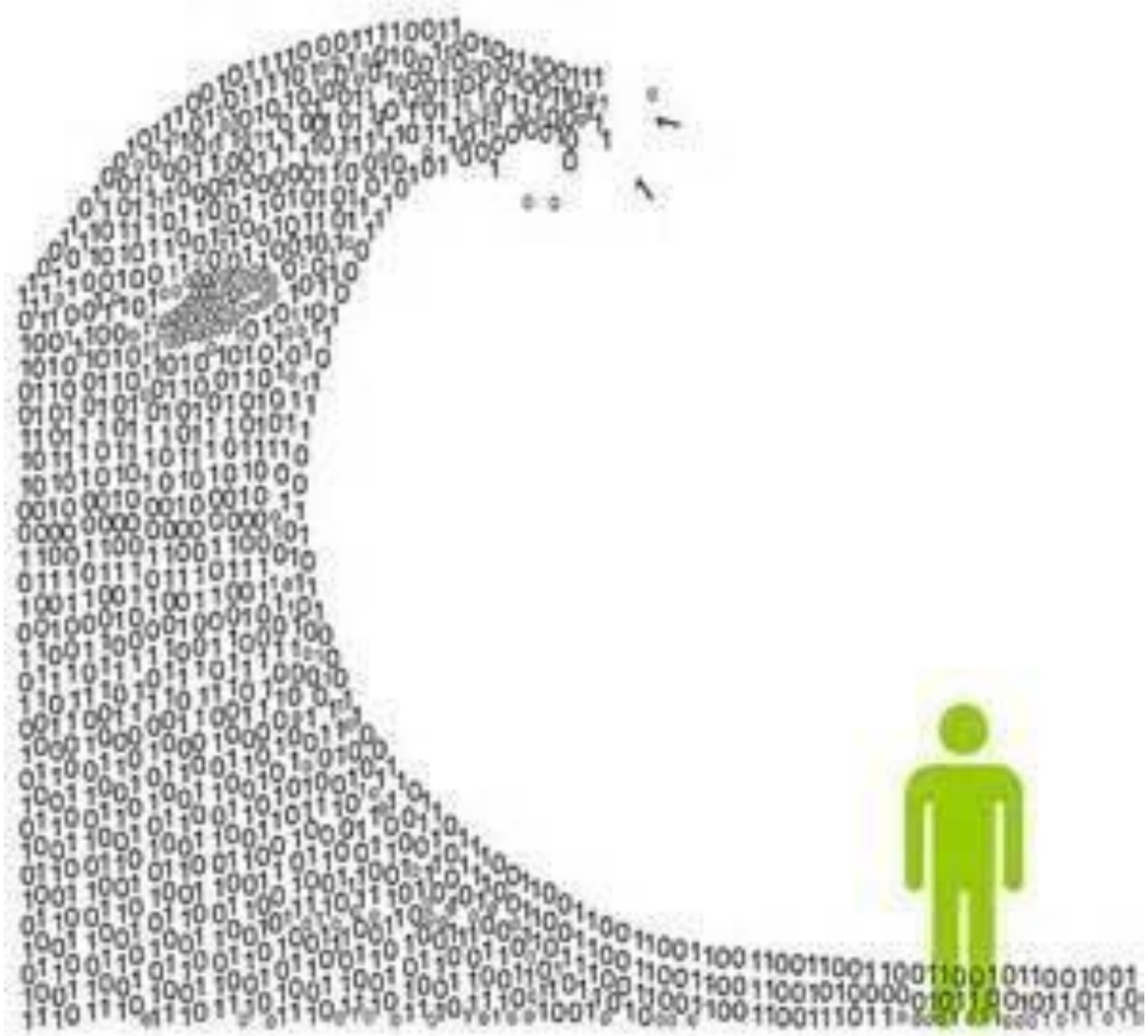

Formato de archivos usados en NGS

Formato FASTA (*FAST All*)

Identificador →
Secuencia →

```
>gi|255767013|ref|NC_000964.3| Bacillus subtilis subsp. subtilis str. 168
ATCTTTTTCGGCTTTTTTTAGTATCCACAGAGGTTATCGACAACATTTTCACATTACCAACCCCTGTGGA
CAAGGTTTTTTCAACAGGTTGTCCGCTTTGTGGATAAGATTGTGACAACCATTGCAAGCTCTCGTTTATT
TTGGTATTATATTTGTGTTTTAACTCTTGATTACTAATCCTACCTTTCCTCTTTATCCACAAAGTGTGGA
TAAGTTGTGGATTGATTTCACACAGCTTGTGTAGAAGGTTGTCCACAAGTTGTGAAATTTGTCGAAAAGC
TATTTATCTACTATATTATATGTTTTCAACATTTAATGTGTACGAATGGTAAGCGCCATTTGCTCTTTTT
TTGTGTTCTATAACAGAGAAAGACGCCATTTTCTAAGAAAAGGAGGGACGTGCCGGAAGATGGAAAATAT
ATTAGACCTGTGGAACCAAGCCCTTGCTCAAATCGAAAAAAAGTTGAGCAAACCGAGTTTTGAGACTTGG
ATGAAGTCAACCAAAGCCCACTCACTGCAAGGCGATACATTAACAATCACGGCTCCCAATGAATTTGCCA
GAGACTGGCTGGAGTCCAGATACTTGCATCTGATTGCAGATACTATATATGAATTAACCGGGGAAGAATT
GAGCATTAAGTTTGTCATTCCTCAAAATCAAGATGTTGAGGACTTTATGCCGAAACCGCAAGTCAAAAAA
```

Recomendado: todas las líneas inferiores a 80 caracteres
Un valor muy común son los 60 caracteres

El genoma de referencia suele presentarse en este formato

Varias extensiones => .fasta .fa .faa .fna

Formato FASTQ (FASTA con calidades)

Bloques de 4 líneas

@Nombre_de_la_lectura
ATCGCCGT…..
+Nombre_de_la_lectura(opcional)
EFGHHGGH…

Las lecturas (*reads*) suelen presentarse en este formato

Formato FASTQ (FASTA con calidades)

Bloques de 4 líneas

```
@EAS20_8_6_88_1477_304/1
CTGATGCGACGCTTGCGCGTCTTATCAGGCCTACAAATTCCCGCACCCTCCGTAGGCCGGATAAGGCGTTTACGCCGCATCCGGCAACAGTGCCGACTGC
+
HHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHFFHHGHIHHHHHHHDEGHEHHGHGGHHBFDGHCEHGEFGGGCGAFG@DEDFGGCEG;CGGE
@EAS20_8_6_88_1477_1713/1
AACTGCTGATGACTGAGTGTGAAATGGTTAACGGTTTTATCGACCCGCCGGGTGAGCCGCCGCACTTCACGCGCGGCTACGGGCTGGTGTTCGGCATGGG
+
IGHHHGHHHHHHHHHHHHHHHHHHHHHHHHHHHHGHHHHHHGGHHGHHGHHHEHHHHIIHDH@CFHGHGGHEBBGGGGFG>HEG5@@##############
@EAS20_8_6_88_1477_1556/1
GCTATGATAACCGTACCAATTATGACGCGTATTATTCTCCCGGTTCACCGTTGCTCGATACCCGTAAACTCTATAGCCAAAGTTGGGACGCCGGGCTGCG
+
EFHHHIHHHHDHHEGHEGEFFAGF=GCAF>EFBFHHHHH@GHFGDHHDEHDF?FD<CBEEHDGGEFBFAC6D<;;EHD??>G.GEGEDE?A<=>?5FFEF
@EAS20_8_6_88_1477_482/1
CGCTTGTTCGGAGGATAAGTTATGGCATTACTCGATTTCTTTCTCTCGCGGAAGAAAAACACAGCCAACATTGCAAAAGAACGGCTGCAGATTATTGTTG
+
HHHHHHGHHHGGHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHGGHHHHHEHHHHHHHHEGGGGHIEHHHHHHGHHHGHBH@?CHHHHHFHH-AFBFHEAHD
@EAS20_8_6_88_1477_1357/1
GGTATGATAGCTGCACCAGGCGCTCATTTGCTGATAAAGCAGTGAAGAGTACAGTTCCAGGTTCATCTGCTCATTAAGTTTTTCAATCATTTCTGGTTTC
+
HGHHGHHHHGGHGHHGHHHHHHGHHIHHHHHHHHHFHHHGHHFHGHGHHGHHHHIHHHGFHGHGGHIFHIHGHIHDBHGGGHHIEBHDCHHGCHE9?FF#
```

Las lecturas (*reads*) suelen presentarse en este formato

Formato GTF/GFF3 (General Transfer Format/General Feature Format)

Consiste en una línea por cada rasgo, cada una conteniendo 9 columnas de datos.
Opcionalmente puede haber una columna adicional con más información.

Mas información en:
http://www.ensembl.org/info/website/upload/gff.html

1. seqname
2. source
3. feature
4. start
5. end
6. score
7. strand
8. frame
9. attribute

```
X       Ensembl Repeat  2419108 2419128 42      .       .       hid=trf; hstart=1; hend=21
X       Ensembl Repeat  2419108 2419410 2502    -       .       hid=AluSx; hstart=1; hend=303
X       Ensembl Repeat  2419108 2419128 0       .       .       hid=dust; hstart=2419108; hend=2419128
X       Ensembl Pred.trans.     2416676 2418760 450.19  -       2       genscan=GENSCAN00000019335
X       Ensembl Variation       2413425 2413425 .       +       .
X       Ensembl Variation       2413805 2413805 .       +       .
```

La anotación de un genoma suele presentarse en este formato

Formato SAM (*Sequence Alignment/Map*)

```
NS500387:17:HC3T3BGXX:4:11403:4287:9101 99      chr1    12024   255     73M     =       12126   176     GCAACTGCTGGCCTGTGCCAGGGTGCAAGCTGAGCACTGGAGTGGAGTTTTCCTGTGGAGAGGAGCCATGCCT       AAA7AFFFFFFFFFFFFFFFFFFFFFF.FFFF.FF)FFFFFFFFFFFFFFF7FFFAFFFFFFFFFFF<FFFF<       XA:i:0  MD:Z:73 NM:i:0
NS500387:17:HC3T3BGXX:3:13404:20726:5062 83      chr1    12026   255     76M     =       11935   -167    AACTGCTGGCCTGTGCCAGGGTGCAAGCTGAGCACTGGAGTGGAGTTTCCCTGTGGAGAGGAGCCATGCCTAGAGT     FFAAFFFFFFFFFAFFFFF)F.F)FFFFFAFFFFFFFAFF)FAFF.FAAFFAFFFFFFFF<A.FFF<FFFFAAAAA    XA:i:1  MD:Z:48T27      NM:i:1
NS500387:17:HC3T3BGXX:2:22110:8289:17152 99      chr1    12045   255     76M     =       12126   155     GGTGCAAGCTGAGCACTGGAGTGGAGTTTTCCTGTGGAGAGGAGCCATGCCTAGAGTGGGATGGGCCATTGTTCAT     A<AAAFFF<.<FFF<FFF<F.AFFFFF7FAFFFF<7F.<AF.F)F.)><FF.7AAF<)).<F<FFF.A.F.FFFAF    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:1:11211:22651:5063 147     chr1    12087   255     76M     =       11986   -177    AGCCATGCCTAGAGTGGGATGGGCCATTGTTCATCTTCTGGCCCCTGTTGTCTGCATGTAACTTAATACCACAACC     <FF<FFFFF<FFFF7FFFFFFAFFFFFFFFFFFFFFFFFAFFFFFFFFFFFFFAFAFFFFFFFFFFFFFFFFFFAAAAA    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:1:12212:10484:3785 147     chr1    12087   255     76M     =       12021   -142    AGCCTTGCCTAGAGTGGTATGGGCCATTGTTCATCTTCTGGCCCCTGTTGTCTGCATGTAACTTAATACCACAACC     FFFF.<FFF.FF7F7FF)F).AFFF.)F..FFFA777.FFFAFFFFFFFAFAFFAAFAFAF77A.A7.A7<FA<AA<    XA:i:2  MD:Z:4A12G58    NM:i:2
NS500387:17:HC3T3BGXX:2:13109:25791:5612 147     chr1    12091   255     76M     =       11970   -197    ATGCCTAGAGTGGGATGGGCCATTGTTCATCTTCTGGCCCCTGTTGTCTGCATGTAACTTAATACCACAACCAGGC     AFFFFFFFFF.AFFF<FFFFFFFFF<)AFFFFFFFFFFFFFFAF<FFFFFFAFFF<AFFFFFFFFFFFFFFFAAAAA    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:2:22110:8289:17152 147     chr1    12126   255     74M     =       12045   -155    GGCCCCTGTTGTCTGCATGTAACTTAATACCACAACCAGGCATAGGGGAAAGATTGGAGGAAAGATGAGTGAGA       ))F..<FF.FFFFFF.FF7AFF<AAF..F<AFFFF<.F..FFAFF).FF<F<F<FFFFF).FFFF)FFF<AAAA    XA:i:0  MD:Z:74 NM:i:0
NS500387:17:HC3T3BGXX:4:11403:4287:9101 147     chr1    12126   255     74M     =       12024   -176    GGCCCCTGTTGTCTGCATGTAACTTAATACCACAACCAGGCATAGGGGAAAGATTGGAGGAAAGATGAGTGAGA       FFFFFFFF7FF)FFFFFFFFFFF<FFFFFFFFFFFFFFFFFFF.FFFFFFFFFFFFFFFFFFFFFFF.AAAA      XA:i:0  MD:Z:74 NM:i:0
NS500387:17:HC3T3BGXX:3:21602:14486:16535 99     chr1    12153   255     74M     =       12181   104     TACCACAACCAGGCATAGGGGAAAGATTGGAGGAAAGATGAGTGAGAGCATCAACTTCTCTCACAACCTAGGCC       AAAAAFF.FFFFFFAF7FFFFFFFFFFFFFFFFFFFFFF.FAF<F.FF.FFFFFFFFFFFF7FFAFFFFFFFF     XA:i:0  MD:Z:74 NM:i:0
NS500387:17:HC3T3BGXX:3:11404:12305:12864:18092 163     chr1    12160   255     74M     =       12251   163     ACCAGGCATAGGGGAAAGATTGGAGGAAAGATGAGTGACAGCATCAACTTCTCTCACAACCTAGGCCAGTAAGT       <AAAAFFFFAFFFFF<FFFFFFFFFFFFFFFFFFFFFFFFFFFF.FFFFFFFF.FAFFFFFFFFFFFFFFF      XA:i:1  MD:Z:38G35      NM:i:1
NS500387:17:HC3T3BGXX:1:22107:13165:6789 99      chr1    12171   255     76M     =       12259   162     GGGAAAGATAGGAGGAAAGATGAGTGACAGCATCACCTTCTCTCACAACCTAGGCCAGTAAGTAGTGCTTGTGCTC     <A7AAAAAF)F.FFFFFAFFFFFFFF7F7FFAFF7.FF)FFFFFA)F)FF<.FFAFFA)FFF.7AA7F<FAFFFF7     XA:i:3  MD:Z:9T17G7A40  NM:i:3
NS500387:17:HC3T3BGXX:3:21602:14486:16535 147     chr1    12181   255     76M     =       12153   -104    GGAGGAAAGATGAGTGAGAGCATCAACTTCTCTCACAACCTAGGCCAGTAAGTAGTGCTTGTGCTCATCTCCTTGG     FFFFFFFFFFFFFF<FFFFFAF<FFFFFFFAFFFAFFAFFFFFFFFFFFFFFFFFFFFFFFFFA7FF.FFFAAAAA    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:4:11410:3717:5609 163     chr1    12200   255     76M     =       12291   165     GCATCAACTTCTCTCACAACATAGGCCAGTAAGTAGTGCTTGTGCTCATCTCCTTGGCTGTGATACGTGGCCGGCCCTG    AA.AAFFFFFFFF7FFFF.F)FF.FFFFFFF.FF.AFFFA<.FFF.F7.<FFA7FAFFFF<FFAFF7A7FFFAFAF    XA:i:1  MD:Z:20C55      NM:i:1
NS500387:17:HC3T3BGXX:4:13404:2856:7784 163     chr1    12204   255     76M     =       12321   193     CAACTTCTCTCACAACCTAGGCCAGTAAGTAGTGCTTGTGCTCATCTCCTTGGCTGTGATACGTGGCCGGCCCTCG    AA<AAFFFFFFAAFAFFFFFFAFAFFFFFFFFFFFFF.FFFF7FFFFFFFFA.FFFA.7FFFFFFFF<FFFFFFFF    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:4:13610:22578:18748 163     chr1    12214   255     76M     =       12323   185     CACAACCTAGGCCAGTAAGTAGTGCTTGTGCTCATCTCCTTGGCTGTGATACGTGGCCGGCCCTCGCTCCAGCAGC     AAA<AFAFFFFFFFFFAFFFFFAAFFFFFFFFFFFFFFFF><FFFFFAAAFF.F7.FFFAFF.<F)F)FFF><FFFFF    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:3:21505:21629:11576 99      chr1    12228   255     76M     =       12282   130     GTAAGTAGTGCTTGTGCTCATCTCCTTGGCTGTGATACGTGGCCGGCCCTCGCTCCAGCAGCTGGACCCCTACCTG     AAAAAFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF<FF)FFFFF7AFFFF<FFFF    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:1:22305:12864:18092 83      chr1    12251   255     72M     =       12160   -163    CCTTGGCTGTGATACGTGGCCGGCCCTCGCTCCAGCAGCTGGACCCCTACCTGCCGTCTGCTGCCATCGGAG        FFF<FAF.F.FFFFFF.FFFFFFFFF.FFFFFFFFFFFF7.FFFFFF7FFFFFFFF7FFFFFFFFFAAAAA     XA:i:0  MD:Z:72 NM:i:0
NS500387:17:HC3T3BGXX:1:23109:20405:4220 99      chr1    12259   255     76M     =       12330   147     GTGATACGTGGCCGGCCCTCGCTCCAGCAGCTGGACCCCTACCTGCCGTCTGCTGCCATCGGAGCCCAACGCCGGG     AAAAAFFF<FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF7FFF.FFFFFF<FFFFFF.<FAAFF    XA:i:1  MD:Z:69A6       NM:i:1
NS500387:17:HC3T3BGXX:1:22107:13165:6789 147     chr1    12259   255     74M     =       12171   -162    GTGATACGTGGCCGGCCCTCGCTCCAGCAGCTGGACCCCTACCTGCCGTCTGCTGCCATCGGAGCCCAAAGCCG       FAFF.FFFFFFFFFF><F.FFF.FFFFFFFF7FFFFFFF<7FF<FFFFFFFFFFFFFA<FFFFFFFF7FAAAAA     XA:i:0  MD:Z:74 NM:i:0
NS500387:17:HC3T3BGXX:3:21505:21629:11576 147     chr1    12282   255     76M     =       12228   -130    CCAGCAGCTGGACCCCTACCTGCCGTCTGCTGCCATCGGAGCCCAAAGCCGGGCTGTGACTGCTCAGACCAGCCGG     FFFFFFFF.<FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFAFFFFFAFFAF7FFFFFFFFFFFFFFAAAAA    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:4:11410:3717:5609 83      chr1    12291   255     74M     =       12200   -165    GGACCCCTACCTGCCGTCTGCTGCCATCGGAGCCCAAAGCCGGGCTGTGACTGCTCAGACCAGCCGGCTGGAGG       .FFFAFFF.FFAFFFFAAFFFFA<AFF)F<FFFF.FFFFAAFFFFFAF)FFAFFFFFFF<FAFFFF.FF)AAAAA     XA:i:0  MD:Z:74 NM:i:0
NS500387:17:HC3T3BGXX:1:12111:13557:6010 99      chr1    12316   255     76M     =       12407   166     ATCGGAGCCCAAAGCCGGGCTGTGACTGCTCAGACCAGCCGGCTGGAGGGAGGGGCTCAGCAGGTCTGGCTTTGGC    <AAAAFFFFFAFFFFFFFFFFFFFFFFFFFFF.FFFFFFFFFAFFFFFFFFFFFFFAFAFFFFFFFFFFFFFFFF    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:4:13404:2856:7784 83      chr1    12321   255     76M     =       12204   -193    AGCCCAAAGCCGGGCTGTGACTGCTCAGACCAGCCGGCTGGAGGGAGGGGCTCAGCAGGTCTAGCTTTGGCCCTGG    FF777F.F77<AFAF.FFFFFFF<FFFFFA<FF<)FFF7FFFFFAFF<FFF.F<FFFFFFFAFFAFFA<FFAAAAA    XA:i:1  MD:Z:62G13      NM:i:1
NS500387:17:HC3T3BGXX:4:13610:22578:18748 83      chr1    12323   255     76M     =       12214   -185    CCCAAAGCCGGGCTGTGACTGCTCAGACCAGCCGGCTGGAGGGAGGGGCGCAGCAGGTCTAGCTTTGGCCCTGGGA    FFF<.FFFFF7FF<FAFFFFFF7FAF.FFFFFFFFFFAFFAFFFFF7FF<7FFFFFFF)FFAFFFAFFFFFFAAAAA    XA:i:2  MD:Z:49T10G15   NM:i:2
NS500387:17:HC3T3BGXX:2:11308:18784:2576 99      chr1    12325   255     73M     =       12426   176     CAAAGCCGGGCTGTGACTGCTCAGACCAGCCGGCTGGAGGGAGGGGCTCAGCAGGTCTGGCTTTGGCCCTGGG       AAA<A)FFFFF..FF)F)FFFF.F.FAFFFFFFF<FFF<FFFF<FFF.FFFFAFFFFF)FAF.F)FAFFF      XA:i:0  MD:Z:73 NM:i:0
NS500387:17:HC3T3BGXX:1:23109:20405:4220 147     chr1    12330   255     76M     =       12259   -147    CCGGGCTGTGACTGCTCAGACCAGCCGGCTGGAGGGAGGGGCTCAGCAGGTCTGGCTTTGGCCCTGGGAGAGCAGG     FFFFFFFFFAFFFFFF7FFFFFF7FFFFFFFFFFFFFFFFFFFF.FFFFFFFFFFFFFFFFFFFFFFFFFFAAAAA    XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:1:22107:7184:3952 99      chr1    12404   255     74M     =       12495   166     GGTGGAAGATCAGGCAGGCCATCGCTGCCACAGAACCCAGTGGATTGGCCTAGGTGGGATCTCTGCGCTCAACA       AAAAAFFFFFF<FFFFFFFF.FFFFFFFF.FFFA.F<FFFFFFFFFFFFFFFFFFFFFFFFFFF.FFFFF.F7     XA:i:1  MD:Z:65A8       NM:i:1
NS500387:17:HC3T3BGXX:1:12111:13557:6010 147     chr1    12407   255     75M     =       12316   -166    GGAAGATCAGGCAGGCCATCGCTGCCACAGAACCCAGTGGATTGGCCTAGGTGGGATCTCTGAGCTCAACAAGCC     FFFFFA.FFFFFFFFFF<FFA.FFAFFFFFFFFFFF.FFFFFFFAFFFFF.FFFF7FFFFFFFFFFFFFAAAAA    XA:i:0  MD:Z:75 NM:i:0
NS500387:17:HC3T3BGXX:2:11308:18784:2576 147     chr1    12426   255     75M     =       12325   -176    CGCTGCCACAGAACCCAGTGGATTGGCCTAGGTGGGATCTCTGAGCTCAACAAGCCCTCTCTGGGTGGTAGGTGC     F7F.FFFFFAF<FFF<FFAFAF.AFFFFFFFA.FF.F)FFFFFFFFFFFFF)FFFAFFFFFFF.F7F7<FAA.AA    XA:i:0  MD:Z:75 NM:i:0
NS500387:17:HC3T3BGXX:1:12105:8367:6392 99      chr1    12470   255     75M     =       12516   122     GCTCAACAAGCCCTCTCTGGGTGGTAGGTGCAGAGACGGGAGGGGCAGAGCCGCAGGCACAGCCAAGAGGGCTGA     AAAAAFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF     XA:i:0  MD:Z:75 NM:i:0
NS500387:17:HC3T3BGXX:1:22107:7184:3952 147     chr1    12495   255     75M     =       12404   -166    AGGTGCAGAGACGGGAGGGGCAGAGCCGCAGGCACAGCCAAGAGGGCTGAAGAAATGGTAGAACGGAGCAGCTGG     )FF7FFFFFFAFFAFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF     XA:i:0  MD:Z:75 NM:i:0
NS500387:17:HC3T3BGXX:1:12105:8367:6392 147     chr1    12516   255     76M     =       12470   -122    AGAGCCGCAGGCACAGCCAAGAGGGCTGAAGAAATGGTAGAACGGAGCAGCTGGTGATGTGTGGGCCCACCGGCCC     FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF     XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:3:22506:15275:7421 99      chr1    12608   255     76M     =       12701   169     CCCAGGTGTGTGGTGATGCCAGGCATGCCCTTCCCCAGCATCAGGTCTCCAGAGCTGCAGAAGACGACGGCCGACT     AAAAAFF<.FFFFFFAFFFFFFFFFFFFFAFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF     XA:i:0  MD:Z:76 NM:i:0
NS500387:17:HC3T3BGXX:2:13208:25864:5279 163     chr1    12664   255     75M     =       12752   164     GCAGAAGACGACGGCCGACTTGGATCACACTCTTGTGAGTGTCCCCAGTGTTGCAGAGGTGAGAGGAGAGTAGAC     AAAAAFF7FF<FFFAFF.FFFFFAFFAFAFFFFFFFFFFFFFFAFFAFFFFFFFFFFFFFFFFFFFFFFFAF     XA:i:0  MD:Z:75 NM:i:0
NS500387:17:HC3T3BGXX:2:13208:25864:5279 163     chr1    12691   255     76M     =       12777   162     CACTCTTGTGAGTGTCCCCAGTGTTGCAGAGGTGAGAGGAGAGTAGACAGTGAGTGGGAGTGGCGTCGCCCCTAGG     AAAAAAFFFFAFFAFF7<FF
```

Formato SAM (*Sequence Alignment/Map*)

```
7001450:326:CAHFUANXX:1:1114:3663:59924 337     1       3151325 0       76M     11      49445610        0       GAAATATCTCTCTGGGCCTTATAGCAGGAGTACTCTGTTACCTTTT
GTGTCTTGTCTAATGTCCGGTGCACCAATC    FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF<BB<<<    AS:i:0  XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i
:0  MD:Z:76 YT:Z:UU NH:i:20 CC:Z:X  CP:i:29734849   HI:i:4
7001450:326:CAHFUANXX:1:2307:3750:36899 369     1       3151462 0       76M     5       59955904        0       TTGTGTTGGATAATAAAGATGGTATAAAAAACTTTATCTGCAAAGC
CGAGAGCTGCCACGTGTTTCAGCCAGGAAT    FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFBBBB0    AS:i:0  XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i
:0  MD:Z:76 YT:Z:UU NH:i:20 CC:Z:14 CP:i:42140262   HI:i:13
7001450:326:CAHFUANXX:1:1210:14912:26804        83      1       3152139 0       76M     10      83403244        0       GGTTATAGGTATTATCCTAACCTTGCACAAAAAACTTA
AAAATTATGGTTAAAACTGCTATTGTTCCATTGACTGC    BBBBBFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF    AS:i:-5 XN:i:0  XM:i:1  XO:i:0  XG:i
:0  NM:i:1  MD:Z:21C54      YT:Z:UU NH:i:20 CC:Z:17 CP:i:13164779   HI:i:10
7001450:326:CAHFUANXX:1:2116:8547:10912 321     1       3152139 0       76M     10      83403233        0       GGTTATAGGTATTATCCTAACCTTGCACAAAAAACTTAAAAATTAT
GGTTAAAACTGCTATTGTTCCATTGACTGC    BBBBBFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFBFFFFF    AS:i:0  XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i
:0  MD:Z:76 YT:Z:UU NH:i:20 CC:Z:14 CP:i:18180824   HI:i:10
7001450:326:CAHFUANXX:1:1111:17693:49345        353     1       3152152 0       76M     X       29735844        0       CTCCTAACCTTGCACAAAAAACTTAAAAATTATGGTTA
AAACTGCTATTGTTCCATTGACTGCAGCTTGCAGTTTG    0<<BBFFFFFFFFFFFFFFFFFFFFFFFBFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF<FBFFFFFBFFF    AS:i:-3 XN:i:0  XM:i:1  XO:i:0  XG:i
:0  NM:i:1  MD:Z:0A75       YT:Z:UU NH:i:20 CC:Z:Y  CP:i:12440976   HI:i:17
7001450:326:CAHFUANXX:1:1211:2683:30958 369     1       3153042 0       76M     10      83402820        0       AATTGTTATTAATTGGTAATTAAAAATTGGTTGTGCCCAAAACAAT
TCTTTGGCCAAAAAGAAAACATTATTGTAA    FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFBBBBB    AS:i:0  XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i
:0  MD:Z:76 YT:Z:UU NH:i:20 CC:Z:12 CP:i:76224842   HI:i:10
```

La versión binaria del SAM es el BAM (*Binary Alignment/Map*)
Usa coordenadas basadas en 0

Más información:
http://samtools.github.io/hts-specs/SAMv1.pdf
http://genome.sph.umich.edu/wiki/SAM

Formato BED (*Browser Extensible Data*)

```
chr1   213941196   213942363
chr1   213942363   213943530
chr1   213943530   213944697
chr2   158364697   158365864
chr2   158365864   158367031
chr3   127477031   127478198
chr3   127478198   127479365
chr3   127479365   127480532
chr3   127480532   127481699
```

Los picos se presentan en este formato
3 columnas obligatorias y 9 columnas adicionales optativas
Usa coordenadas basadas en 0

Más información:
http://www.ensembl.org/info/website/upload/bed.html

Formato BED (*Browser Extensible Data*)

chr1	213941196	213942363
chr1	213942363	213943530
chr1	213943530	213944697
chr2	158364697	158365864
chr2	158365864	158367031
chr3	127477031	127478198
chr3	127478198	127479365
chr3	127479365	127480532
chr3	127480532	127481699

1. chrom
2. chromStart
3. chromEnd
4. name
5. score
6. strand
7. thickStart
8. thickEnd
9. itemRgb
10. blockCount
11. blockSizes
12. blockStarts

chr7	127471196	127472363	Pos1	0	+	127471196	127472363	255,0,0
chr7	127472363	127473530	Pos2	0	+	127472363	127473530	255,0,0
chr7	127473530	127474697	Pos3	0	+	127473530	127474697	255,0,0
chr7	127474697	127475864	Pos4	0	+	127474697	127475864	255,0,0
chr7	127475864	127477031	Neg1	0	-	127475864	127477031	0,0,255
chr7	127477031	127478198	Neg2	0	-	127477031	127478198	0,0,255
chr7	127478198	127479365	Neg3	0	-	127478198	127479365	0,0,255
chr7	127479365	127480532	Pos5	0	+	127479365	127480532	255,0,0
chr7	127480532	127481699	Neg4	0	-	127480532	127481699	0,0,255

Formato VCF (*Variant Call Format*)

```
##reference=cs8.fa
##INFO=<ID=DP,Number=1,Type=Integer,Description="Raw Depth">
##INFO=<ID=AF,Number=1,Type=Float,Description="Allele Frequency">
##INFO=<ID=SB,Number=1,Type=Integer,Description="Phred-scaled strand bias at this position">
##INFO=<ID=DP4,Number=4,Type=Integer,Description="Counts for ref-forward bases, ref-reverse, alt-forward and alt-reverse bases">
##INFO=<ID=INDEL,Number=0,Type=Flag,Description="Indicates that the variant is an INDEL.">
##INFO=<ID=CONSVAR,Number=0,Type=Flag,Description="Indicates that the variant is a consensus variant (as opposed to a low frequency variant).">
##INFO=<ID=HRUN,Number=1,Type=Integer,Description="Homopolymer length to the right of report indel position">
##FILTER=<ID=min_dp_10,Description="Minimum Coverage 10">
##FILTER=<ID=sb_fdr,Description="Strand-Bias Multiple Testing Correction: fdr corr. pvalue > 0.001000">
##FILTER=<ID=min_snvqual_51,Description="Minimum SNV Quality (Phred) 51">
##FILTER=<ID=min_indelqual_20,Description="Minimum Indel Quality (Phred) 20">
#CHROM  POS     ID      REF     ALT     QUAL    FILTER  INFO
cs8     2858    .       C       T       474     PASS    DP=536458;AF=0.000746;SB=0;DP4=535951,0,400,0
cs8     2861    .       T       G       757     PASS    DP=536545;AF=0.001239;SB=0;DP4=535644,0,665,0
cs8     2862    .       C       T       394     PASS    DP=536560;AF=0.000535;SB=0;DP4=536100,0,287,0
cs8     2866    .       C       T       78      PASS    DP=536575;AF=0.000397;SB=0;DP4=536303,0,213,0
cs8     2872    .       T       G       56      PASS    DP=536589;AF=0.000593;SB=0;DP4=536098,0,318,0
cs8     2895    .       C       A       62      PASS    DP=536632;AF=0.000669;SB=0;DP4=536180,0,359,0
cs8     2897    .       C       T       49314   PASS    DP=536632;AF=0.988758;SB=0;DP4=5820,0,530599,0
cs8     2904    .       T       C       123     PASS    DP=536636;AF=0.001377;SB=0;DP4=535277,0,739,0
cs8     2915    .       T       C       92      PASS    DP=536595;AF=0.000518;SB=0;DP4=536128,0,278,0
cs8     2917    .       G       T       150     PASS    DP=536595;AF=0.000850;SB=0;DP4=536049,0,456,0
cs8     2920    .       T       G       494     PASS    DP=536595;AF=0.000947;SB=0;DP4=535902,0,508,0
cs8     2934    .       C       A       52      PASS    DP=536644;AF=0.000617;SB=15;DP4=536100,48,330,1
cs8     2936    .       T       G       645     PASS    DP=536644;AF=0.001371;SB=26;DP4=535730,48,734,2
cs8     2937    .       G       T       143     PASS    DP=536644;AF=0.000826;SB=0;DP4=535946,43,443,0
cs8     2943    .       A       C       766     PASS    DP=536644;AF=0.001552;SB=25;DP4=535527,48,831,2
cs8     2944    .       C       A       71      PASS    DP=536644;AF=0.000883;SB=0;DP4=536045,47,474,0
cs8     2946    .       A       C       177     PASS    DP=536644;AF=0.001276;SB=12;DP4=535581,47,684,1
cs8     2969    .       A       C       263     PASS    DP=536639;AF=0.001137;SB=0;DP4=535814,49,610,0
cs8     2978    .       A       C       222     PASS    DP=536639;AF=0.001399;SB=11;DP4=535591,48,750,1
cs8     2982    .       A       C       209     PASS    DP=536638;AF=0.001699;SB=11;DP4=535275,47,911,1
cs8     2987    .       G       T       764     PASS    DP=536638;AF=0.001744;SB=10;DP4=535433,49,935,1
cs8     2990    .       A       C       1093    PASS    DP=536638;AF=0.002417;SB=0;DP4=534902,50,1297,0
cs8     2992    .       A       C       57      PASS    DP=536638;AF=0.001653;SB=24;DP4=535346,48,885,2
cs8     2995    .       G       T       1069    PASS    DP=536638;AF=0.002177;SB=0;DP4=535220,49,1168,0
cs8     2996    .       G       T       63      PASS    DP=536637;AF=0.001310;SB=0;DP4=535661,50,703,0
cs8     2997    .       T       G       97      PASS    DP=536637;AF=0.000891;SB=0;DP4=535725,50,478,0
```

Formato VCF (*Variant Call Format*)

Más información:

The Variant Call Format (VCF) Version 4.2 Specification

24 May 2017

https://samtools.github.io/hts-specs/VCFv4.2.pdf

MUCHAS GRACIAS POR SU ATENCIÓN
¿PREGUNTAS?